

Robust Hybrid Learning for Automatic Teeth Segmentation and Labeling on 3D Dental Models

Shaojie Zhuang^{ID}, Guangshun Wei^{ID}, Zhiming Cui, and Yuanfeng Zhou^{ID}

Abstract—Automatic teeth segmentation and labeling on dental models are basic tasks in computer-aided dentistry. Many existing works can achieve promising results in teeth segmentation, but they heavily rely on aligned input dental models, which leads to additional manual intervention. Moreover, tooth labeling is an essential task in digital dentistry for treatment planning (e.g., orthodontic), and is usually ignored in these methods. In this article, we propose an AlignNet for aligning dental models of arbitrary sizes and orientations automatically. Meanwhile, a multi-task hybrid learning network is designed that effectively plays the advantages of semantic segmentation and instance segmentation, and synergistically improves the performance of teeth point clouds segmentation and labeling. Particularly, for the teeth-gingival boundaries with large segmentation errors, we utilize the filtered curvature information as a constrained feature to detect the weak boundary more accurately. At last, we propose a DiffLoss and postprocessing step based on the dental arch to address the teeth classification problem. Through extensive evaluations of oral scanning models, our method is robust to handle dental model point clouds with arbitrary size and orientation, and outperforms state-of-the-art teeth segmentation and labeling methods, demonstrating its full automation and robustness in clinical practice.

Index Terms—Dental model segmentation, teeth labeling, 3D point cloud.

I. INTRODUCTION

IN RECENT years, with the development of computing technology, computer-aided design systems have attracted much attention in many fields. Most clinics use digital dental CAD systems to assist dentists in different applications, e.g., implants, orthodontics, or dental restorations. At present, clinics or orthodontic hospitals generally use the intra-oral scanning model as the basic data due to its high-precision crown representation. Hence, automatic and accurate teeth segmentation and labeling on digital 3D dental models are crucial steps in digital dentistry.

Manuscript received 14 October 2022; revised 1 April 2023 and 13 May 2023; accepted 13 June 2023. Date of publication 30 June 2023; date of current version 21 February 2025. This work was supported in part by the National Key R&D Plan on Strategic International Scientific and Technological Innovation Cooperation Special Project under Grant 2021YFE0203800, in part by the NSFC-Zhejiang Joint Fund of the Integration of Informatization and Industrialization under Grant U1909210, and in part by the National Natural Science Foundation of China under Grant 62172257. The guest editor coordinating the review of this manuscript and approving it for publication was Prof. Yanwen Guo. (Shaojie Zhuang and Guangshun Wei contributed equally to this work.) (Corresponding author: Yuanfeng Zhou.)

Shaojie Zhuang, Guangshun Wei, and Yuanfeng Zhou are with the School of Software, Shandong University, Jinan 250100, China (e-mail: shjie.zhuang@gmail.com; guangshunwei@gmail.com; yfzhou@sdu.edu.cn).

Zhiming Cui is with the ShanghaiTech University, Shanghai 201210, China (e-mail: cuizhm@shanghaitech.edu.cn).

Digital Object Identifier 10.1109/TMM.2023.3289760

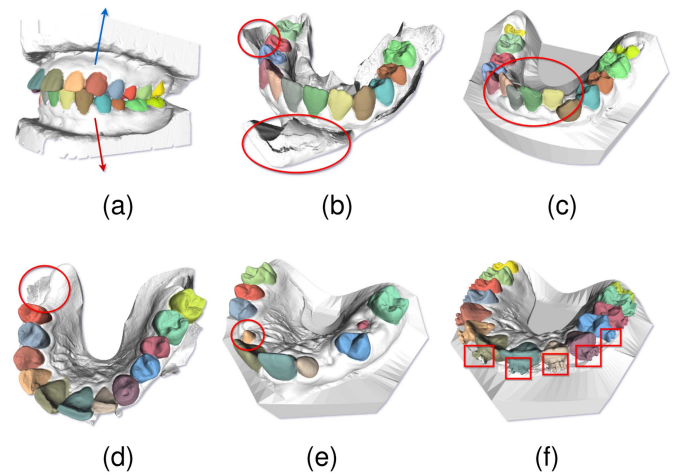


Fig. 1. Representative dental models. (a) Different model poses and sizes. (b) Interfering gingiva parts. (c) Crowded teeth. (d) Missing teeth. (e) Small teeth. (f) Teeth with dental attachments.

However, robust segmentation and labeling of individual teeth from intra-oral scanning data face several steep challenges. First, the digital dental models obtained by different scanning devices usually have different angle and size standards in 3D space (as shown in Fig. 1(a)), where the misaligned crown data seriously drop the accuracy of subsequent segmentation and labeling tasks. Second, there are also great differences in the gingival parts obtained by different devices (as shown in Fig. 1(b) and (c)), and the protruding part of the gingiva will be mistakenly segmented as individual teeth. Third, varying dentition morphology caused by dental abnormalities in orthodontic treatments, including crowded teeth, impacted teeth, missing teeth in the elderly, small teeth in adolescents, and teeth with attachments (as shown in Fig. 1(c)–(f)), which significantly affect the segmentation robustness. Fourth, teeth of the same type have similar shapes and sizes, which will easily cause label ambiguities. It is challenging to consider the relationship between teeth and accurately distinguish their labels.

These difficulties lead to the lack of robustness of traditional segmentation methods based on geometric features defined with fixed parameters. For example, the curvature-based methods [1], [2] segment teeth and gingivae by taking the curvature features of crown shapes as prior knowledge. However, teeth and gingival surfaces usually have many artifacts with non-stable signals that cause serious interference with teeth segmentation. To this end, some semi-automatic algorithms [3], [4] are designed to

improve the robustness and accuracy of teeth segmentation by introducing human-computer interaction strategies.

In recent years, convolutional neural networks [5] and transformers [6] have achieved state-of-the-art results and significantly exceeds previous traditional methods in many applications. Meanwhile, point cloud [7], [8] or mesh-based [9] neural networks have also been proposed for segmentation and classification on 3D data. Zanjani et al. propose a data-driven approach (Mask-MCNet) [10], converting dental models into point cloud data followed by a volume-anchored-based region proposal network, for tooth detection and segmentation. In addition, Cui et al. propose TSegNet [11], which detects all teeth using a distance-aware tooth centroid voting strategy, and then individually segments the teeth based on detected centroids. Unfortunately, all these methods make a strong restrictive assumption that the dental models are aligned with similar gingival distribution, which is impractical in clinics. Moreover, the datasets used in these methods are dental models with complete teeth conditions, while severely missing teeth and small teeth derived from adolescents and the elderly are not carefully considered. Moreover, the teeth labeling task is also extremely important in digital dental systems (e.g., model alignment in orthodontics). However, existing methods focus more on teeth segmentation accuracy and ignore the crucial teeth labeling task.

In this article, we design a teeth segmentation and labeling framework that can robustly and accurately segment and classify individual teeth for digital dental models. Specifically, we first propose a model alignment strategy based on any device-acquired or arbitrary-shaped teeth, which can effectively improve the robustness of downstream teeth segmentation and labeling. Then, we combine the semantic segmentation (point-wise) with instance segmentation (object-level) together for more precise segmentation and labeling results. Specifically, we design a novel DiffLoss for semantic segmentation on dental models to find the optimal label for each tooth instance. Furthermore, we use the dental arch curve as strong prior knowledge to correct potential labeling errors. During the tooth instance segmentation stage, we share a similar idea with TSegNet [11] that firstly predicts the tooth centroids and then segments each tooth. To delineate more accurate crown boundaries, the curvature features, a piece of useful complementary information, are introduced in our method efficiently. Due to the negative curvature characteristics of the gingivae and teeth, we further propose a filtering strategy that employs a crown region binary mask to filter out the curvature features of interference regions. Finally, we conduct extensive experiments on the Teeth3DS [12] dataset. Compared with previous methods, our method achieves state-of-the-art performance on both segmentation and labeling tasks in terms of different evaluation metrics. To sum up, the main contributions of this article are as follows:

- For varying spatial differences of the 3D tooth models obtained by different equipment, we propose an AlignNet based on the transformer mechanism to solve uncertain postures of different models collected in clinical practice.
- We propose a hybrid learning strategy on point-level and tooth-level that effectively combines the advantages of semantic segmentation and instance segmentation with a

novel label optimization strategy, which can synergistically improve the accuracy of teeth segmentation and classification.

- We propose a novel evaluation metric, practical usability, used to check whether all the teeth are correctly labeled in a dental model. This metric has essential reference value for reducing manual intervention in practical applications of digital orthodontics. Our method gained huge improvement compared to other algorithms.
- For different dental abnormalities (e.g., crowded teeth, missing teeth, small teeth, and broken teeth), our algorithms can accurately and robustly obtain high-precision segmentation and labeling results.

II. RELATED WORKS

A. Deep Learning on Point Cloud

In recent years, deep learning methods on point cloud understanding have developed a lot. PointNet [7] directly applies multi-layer perceptions and max-pooling to point clouds, and gains a great performance, but it ignores the local relations of points. Then, many other works like PointNet++ [8], PointCNN [13], DGCNN (EdgeConv) [14], PointConv [15], KPConv [16], DPC [17] and RSCNN [18] take local neighborhood relationship into account, which is highly important for understanding a point cloud. Since self-attention [19] achieved great success in natural language processing, more and more works [6], [20], [21] use Transformer to computer vision tasks, but most of them face a problem of unbearable computational cost. Point Transformer [22] studies self-attention in the local neighborhood and spatial location. Point Cloud Transformer [23] proposes offset attention, which saves a lot of computational resources while keeping high performance. Although these deep learning methods achieve great results on many segmentation and classification tasks, they have to balance between accuracy and computational costs when directly applied to high-precision dental models. Usually, due to the lack of the prior about the position of each tooth, these methods will not distinguish each tooth from others well.

B. Dental Model Segmentation

Interactive methods of dental model segmentation have always been an effective solution. The key point of interactive methods is finding a suitable boundary and making segmentation fields. An important feature at boundary points is curvature. A common way is setting a curvature threshold, manually [24], [25] or automatically [26], and then splitting a whole model into several segmentation fields. Users need to select patches that belong to teeth. These methods depend on high-quality models and human labors, and are not efficient enough.

With the development of deep learning methods on 3D data, more and more deep learning methods are applied to the dental model segmentation task. Tian et al. [27] use sparse voxel octree and 3D convolution neural networks for automatic segmentation, but the hierarchical structure it uses is sensitive to errors at each

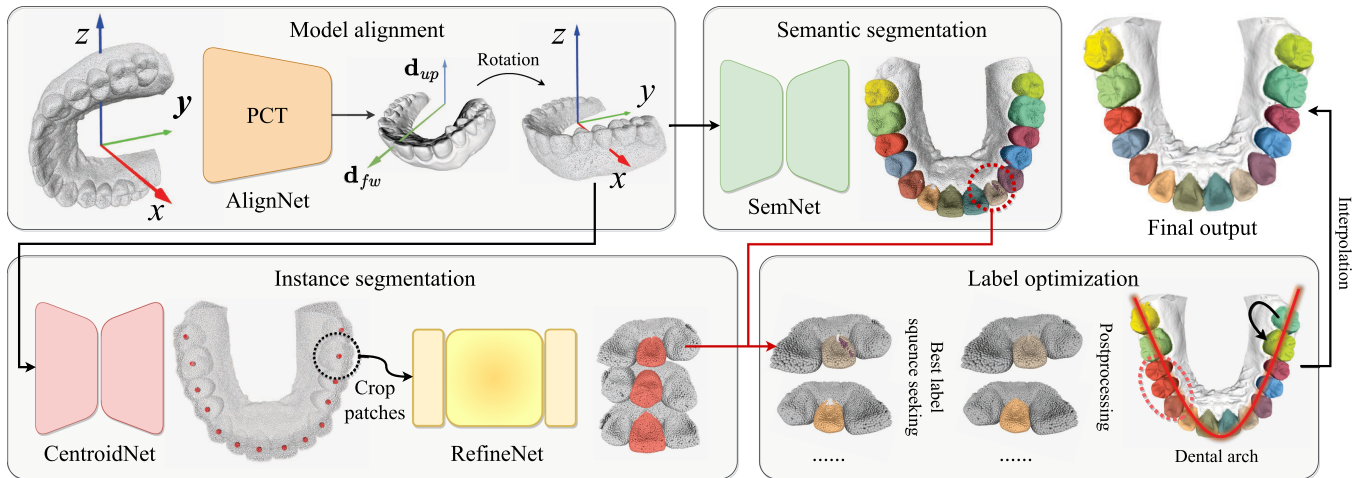


Fig. 2. Pipeline of our method. We first use an AlignNet to align an input intraoral scan model to a unified pose. Next, semantic segmentation is made, where we get possible labels for each tooth and a teeth-gingival segmentation. Then, we predict centroids to detect individual teeth, crop patches around them, and segment one tooth inside each patch. We mask labels from semantic segmentation results by patch segmentation masks and find the best label sequence. Finally, a postprocessing stage is applied in case there exist potential labeling errors. The labels are interpolated to the original dental model and form the final output result.

layer. Another way is using multi-modality data, like X-ray images. Zhao et al. [28], [29] try to segment teeth based on matching 3D dental models and 2D images. It is useful but it requires lots of paired data and cannot handle dental abnormalities like supernumerary teeth well. Some mesh-based methods [9], [30], [31], [32] benefit from topological information provided by the mesh, so they have an advantage in accurately segmenting individual teeth. However, these methods require a fixed face count for each model, which is more demanding than a fixed count of points. Sun et al. [33] add region, boundary, and shape constraints for better segmentation with fewer noises. Recently, Tian et al. [34] combine objectness and pointwise knowledge for point-based segmentation, Liu et al. [35] focus on the functionality of heterogeneous geometry data, and Im et al. [36] combine the traditional curvature-based methods with deep learning methods to improve the segmentation quality at teeth boundaries.

C. Teeth Labeling

Most methods pursue accurate individual teeth segmentation, while teeth labels only occupy an additional output branch of the network. The importance of teeth labeling is always ignored, which blocks the practical application of teeth segmentation algorithms. Lalitha et al. [37] apply graph theories on dental arch for graceful labeling. With the help of teeth feature vectors with neighborhood relations, Ma et al. [38] successfully improves teeth classification accuracy, which most methods do not pay enough attention to. However, these methods consider less global teeth distributions, which will lead to more ambiguity in the classification of similar teeth.

III. METHODS

A. Overview

Commonly used end-to-end multi-category teeth segmentation networks tend to focus more on global features but may

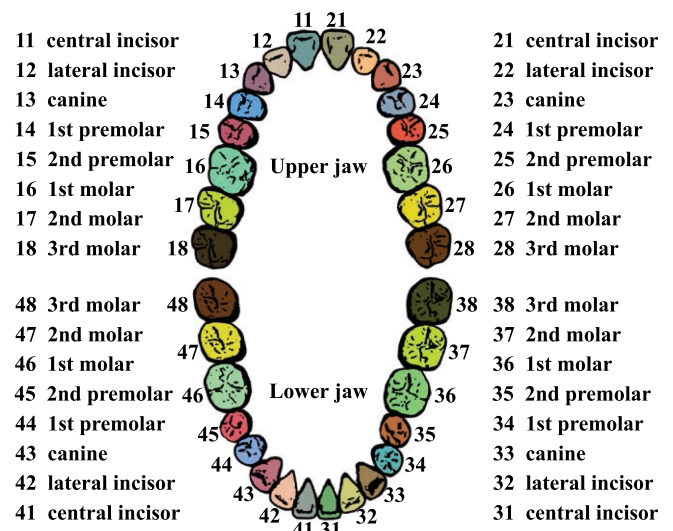


Fig. 3. Fédration Dentaire Internationale (FDI) notation label and color for each tooth.

overlook subtle local features, which leads to more accurate prediction of tooth label distributions, but also more outliers in the segmentation. In this section, we propose a multi-stage framework for teeth segmentation and labeling. As shown in Fig. 2, our method takes a dental model in the form of a point cloud as input and assigns the Fédration Dentaire Internationale (FDI) notation label to each point as output. The FDI label and the color map are illustrated in Fig. 3. Specifically, we first apply a preprocessing operation to the input points that aligns the model to a unified pose. Next, we make a semantic segmentation supervised by proposed DiffLoss. Then, we predict tooth masks in patches cropped around predicted centroids. Finally, we combine the semantic segmentation and instance masks by optimizing the masked labels, and then obtain the final output labels through a postprocessing stage.

Semantic segmentation takes the teeth distribution and their relationships into account and assigns point-wise semantic labels, while instance segmentation masks individual teeth with clear boundaries. We combine their advantages to obtain more accurate segmentation and classification labels than applying semantic or instance segmentation alone.

B. Model Alignment

Due to differences between scanning devices, the orientations of dental models are dramatically varying across devices, which leads to great inconvenience to deep learning methods. Most teeth segmentation and labeling methods are based on manually aligned teeth datasets. Therefore, a proper alignment method could fundamentally improve the robustness of our framework.

Considering non-learning methods like Principal Components Analysis (PCA) [39] or Iterative Closest Point (ICP) [40] are usually sensitive to the model shapes especially with irregular crown positions, we propose a learning-based strategy to align these dental models. First, we define an up direction $\mathbf{d}_{up} = [0, 0, 1]^\top$, and a forward direction $\mathbf{d}_{fw} = [0, -1, 0]^\top$ in world axis as standard directions, as shown in the model alignment section of Fig. 2. The up direction indicates the orientation perpendicular to the occlusal plane, while the forward direction indicates a direction from the root of the tongue to the incisors. In most cases, the upper and lower dental models are usually occluded, *i.e.* they have opposite up directions. However, in some cases, the upper models are flipped vertically, so both upper and lower models share the same up direction. Since our ground truth data are well aligned, we make random rotations to dataset and get new direction vectors $\mathbf{d}'_{up} = \mathbf{R} \cdot \mathbf{d}_{up}$ and $\mathbf{d}'_{fw} = \mathbf{R} \cdot \mathbf{d}_{fw}$, where $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ is a randomly generated rotation matrix. When processing a rotated model, we first use PCA [39] to find the approximate up direction, which minimizes the angle errors compared to the forward direction. Then, with such initial pose, we input the dental model point cloud into a Point Cloud Transformer [23] block for feature extraction in our AlignNet and regress the normalized $\hat{\mathbf{d}}_{up}$ and $\hat{\mathbf{d}}_{fw}$ directly. The loss function is designed as follows:

$$\mathcal{L}_{align} = 2 - \frac{\hat{\mathbf{d}}_{up} \cdot \mathbf{d}'_{up}}{\|\hat{\mathbf{d}}_{up}\| \|\mathbf{d}'_{up}\|} - \frac{\hat{\mathbf{d}}_{fw} \cdot \mathbf{d}'_{fw}}{\|\hat{\mathbf{d}}_{fw}\| \|\mathbf{d}'_{fw}\|} + \hat{\mathbf{d}}_{up} \cdot \hat{\mathbf{d}}_{fw}, \quad (1)$$

where the $\mathcal{L}_{align}$ supervises the angles between predicted vectors and ground truth vectors by cosine similarity, and ensures $\hat{\mathbf{d}}_{up}$ and $\hat{\mathbf{d}}_{fw}$ are orthogonal by dot product. After prediction, we get rotation matrices from $\langle \hat{\mathbf{d}}_{up}, \hat{\mathbf{d}}_{fw} \rangle$ to $\langle \mathbf{d}_{up}, \mathbf{d}_{fw} \rangle$ using Euler—Rodrigues formula and apply them to the input data. The following stages are all built on the aligned data.

C. Semantic Segmentation

Semantic segmentation networks can recognize individual teeth and gingiva with point-wise semantic labels, but cannot precisely distinguish each tooth. We make use of this point to get semantic labels and teeth-gingival separation through a SemNet.

As shown in Fig. 4, we use an encoder-decoder structure with a transformer block for semantic segmentation. To better train the model, we propose a novel loss function, named DiffLoss, which ensures that for any two teeth, the closer their distance, the smaller the gap between their labels, and vice versa.

The DiffLoss $\mathcal{L}_{diff}$ is based on the cropped patches. Assume that a dental model has T tooth instances. For the i -th tooth, l_i denotes the ground truth label, and $\hat{l}_i$ denotes the predicted label. The $\hat{l}_i$ is calculated by choosing the predicted label with the most occurrences in the area $\{\mathbf{p} \mid l_{\mathbf{p}} = l_i\}$, where $\mathbf{p}$ is a point from the point cloud $\mathbf{P}$ and $l_{\mathbf{p}}$ is its ground truth label. The detailed formula of $\mathcal{L}_{diff}$ is defined as:

$$\mathcal{L}_{diff} = \frac{1}{T^2} \sum_{i=1}^T \sum_{j=1}^T [D_l(l_i, l_j) - D_l(\hat{l}_i, \hat{l}_j)]^2, \quad (2)$$

where

$$D_l(a, b) = \begin{cases} |a \bmod 10 - b \bmod 10|, & [a/10] = [b/10] \\ a \bmod 10 + b \bmod 10 + 1, & [a/10] \neq [b/10], \end{cases} \quad (3)$$

the function $D_l(a, b)$ calculates the gap between the FDI labels a and b . Since a FDI label l ($l > 0$) is a two-digit number, the $[l/10]$ is the quadrant for the id l and $l \bmod 10$ is the tooth's position on the dental arch. Unlike Euclidean distance, the label distance is independent of model size and shape, so it is more suitable for this task.

Except for the DiffLoss, we add another cross-entropy loss to all the points. Finally, the total loss for semantic segmentation is defined as:

$$\mathcal{L}_{sem} = \mu \mathcal{L}_{diff} - \frac{1}{|\mathbf{P}|} \sum_{\mathbf{p}_i \in \mathbf{P}} \mathbf{l}_{\mathbf{p}_i} \cdot \log \hat{\mathbf{l}}_{\mathbf{p}_i}, \quad (4)$$

where $\mathbf{l}_{\mathbf{p}_i}$ and $\hat{\mathbf{l}}_{\mathbf{p}_i}$ are the one-hot ground truth labels and one-hot predicted labels respectively, and μ is a balance factor which is set to 0.2 after verification.

D. Instance Segmentation

Tooth centroids prediction: Semantic segmentation focuses more on the global features used to distinguish tooth labels, but it is difficult to accurately distinguish the local teeth and gums. Tooth centroids prediction is only used to detect the approximate position of each tooth without the need for accurate segmentation of tooth boundaries.

As shown in Fig. 5, with the help of sampling and grouping operations, the aligned input point cloud $\mathbf{P}$ with N points is down-sampled to 512 points, denoted as $\mathbf{P}'$, and we use multi-layer perceptions (MLPs) to regress per-point centroid offsets $\Delta \hat{\mathbf{C}}$. The predicted centroids are obtained by $\hat{\mathbf{C}} = \mathbf{P}' + \Delta \hat{\mathbf{C}}$, as shown in Fig. 6(a). Let $\mathbf{C}$ denote the ground truth centroids with shape $T \times 3$, where T is the count of tooth instances in the case. The loss function for centroid prediction $\mathcal{L}_{ct}$ is a sum of a chamfer distance loss $\mathcal{L}_{CD}$, an offset loss $\mathcal{L}_{off}$ and a separation loss $\mathcal{L}_{sep}$. It makes each predicted centroid as close as possible to its nearest ground truth centroid while keeping far away from

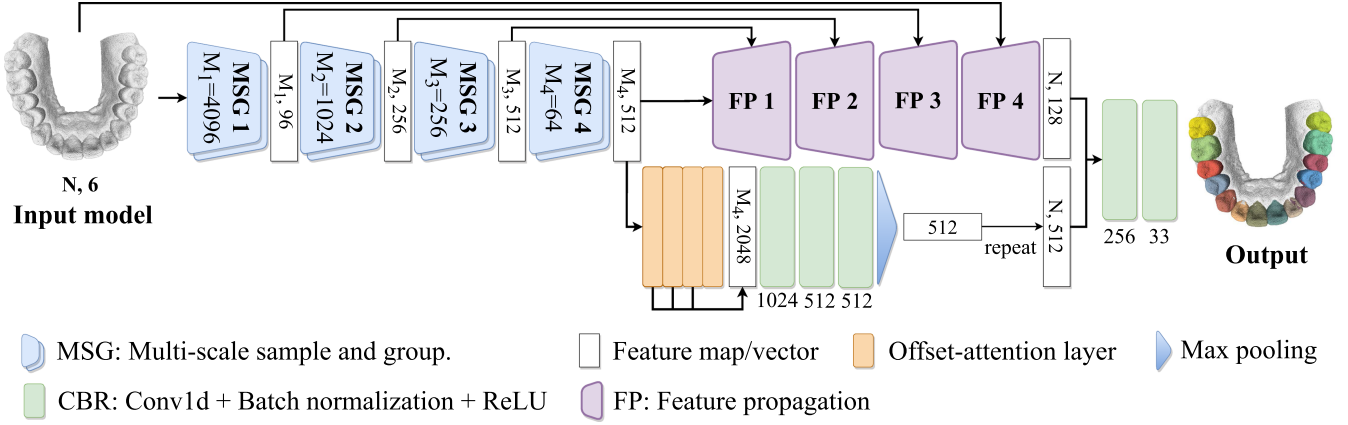


Fig. 4. Network structure of SemNet for semantic segmentation, the output result of which will be used for teeth-gingival separation and patch label determination.

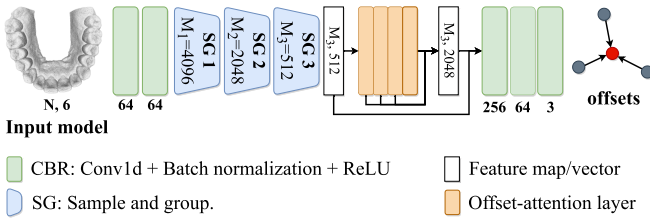


Fig. 5. Network structure of CentroidNet for centroid prediction.

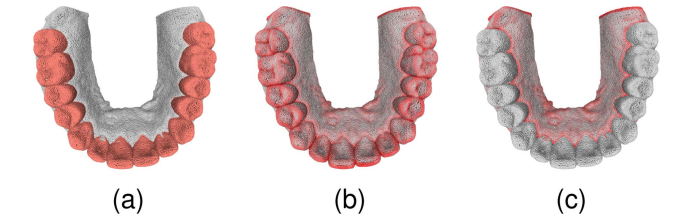


Fig. 7. Teeth-gingival separation. (a) Teeth-gingival binary classification result. (b) Original model curvature. (c) Curvature removed at teeth crown.

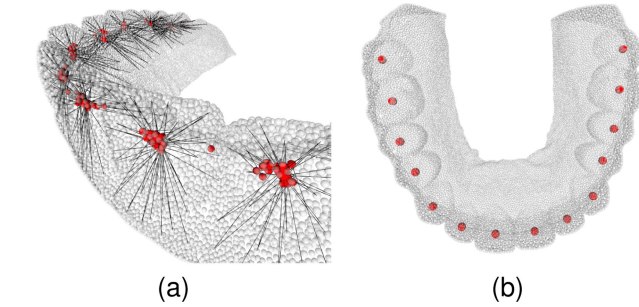


Fig. 6. Centroid prediction result. (a) Dense predicted centroids (red points) from down-sampled points with offset vectors (black lines) in a local area. (b) Clustered centroids.

other centroids, defined as:

$$\mathcal{L}_{CD} = \sum_{\hat{c}_i \in \hat{C}} \min_{c_j \in C} \|\hat{c}_i - c_j\|_2^2 + \sum_{c_j \in C} \min_{\hat{c}_i \in \hat{C}} \|c_j - \hat{c}_i\|_2^2, \quad (5)$$

$$\mathcal{L}_{off} = \sum_i L_1^{smooth}(\|\Delta \hat{c}_i\|_2 - \min_{c_j \in C} \|p'_i - c_j\|_2), \quad (6)$$

$$\mathcal{L}_{sep} = \sum_{\hat{c}_i \in \hat{C}} \frac{\|\hat{c}_i - c_{i1}\|_2}{\|\hat{c}_i - c_{i2}\|_2}, \quad (7)$$

$$\mathcal{L}_{ct} = \mathcal{L}_{CD} + \mathcal{L}_{off} + \lambda \mathcal{L}_{sep}, \quad (8)$$

where $\Delta \hat{c}_i \in \Delta \hat{C}$ is the i -th predicted centroid offset; $p'_i \in P'$ is the i -th down-sampled point; c_{i1} and c_{i2} are the ground truth centroids with the minimum distance and the second minimum

distance from the predicted centroid $\hat{c}_i$ respectively; λ is a coefficient added to $\mathcal{L}_{sep}$, which is set to 0.2, identical to the setting in TSegNet [11].

Too many non-tooth points will interfere with the prediction of centroids of small teeth. Thus, during the inference stage, only the points close to the positive predicted points are sent into the CentroidNet. This operation helps reduce the number of non-tooth points, therefore increasing the proportion of small teeth points, and it helps locate small teeth. Then, we apply the density peaks clustering algorithm [41] to get the predicted centroids, as shown in Fig. 6(b).

Patch-based segmentation with filtered curvature: The curvature information at the teeth-gingival junction is an essential feature to determine the true contour of the tooth more precisely. However, the curvature of the tooth cusps, gingiva, and braces varies greatly, which brings difficulty to traditional curvature-based segmentation methods. Thus, we convert the semantic segmentation into binary classification between foreground teeth and background gingiva (as shown in Fig. 7(a)), and remove curvature on teeth crown (as shown in Fig. 7(b) and (c)). Specifically, we first obtain the curvature information $f_{p_i}^c$ and predicted binary labels $\hat{l}_{p_i}^b$ of each point on the model, where $\hat{l}_{p_i}^b$ is equal to 0 or 1. We represent the filtered curvature feature by defining the following equation:

$$f_{p_i} = \begin{cases} f_{p_i}^c, & \text{if } 0 < \frac{1}{k} \sum_{p_j \in \mathcal{N}(p_i)} \hat{l}_{p_j}^b < 1 \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

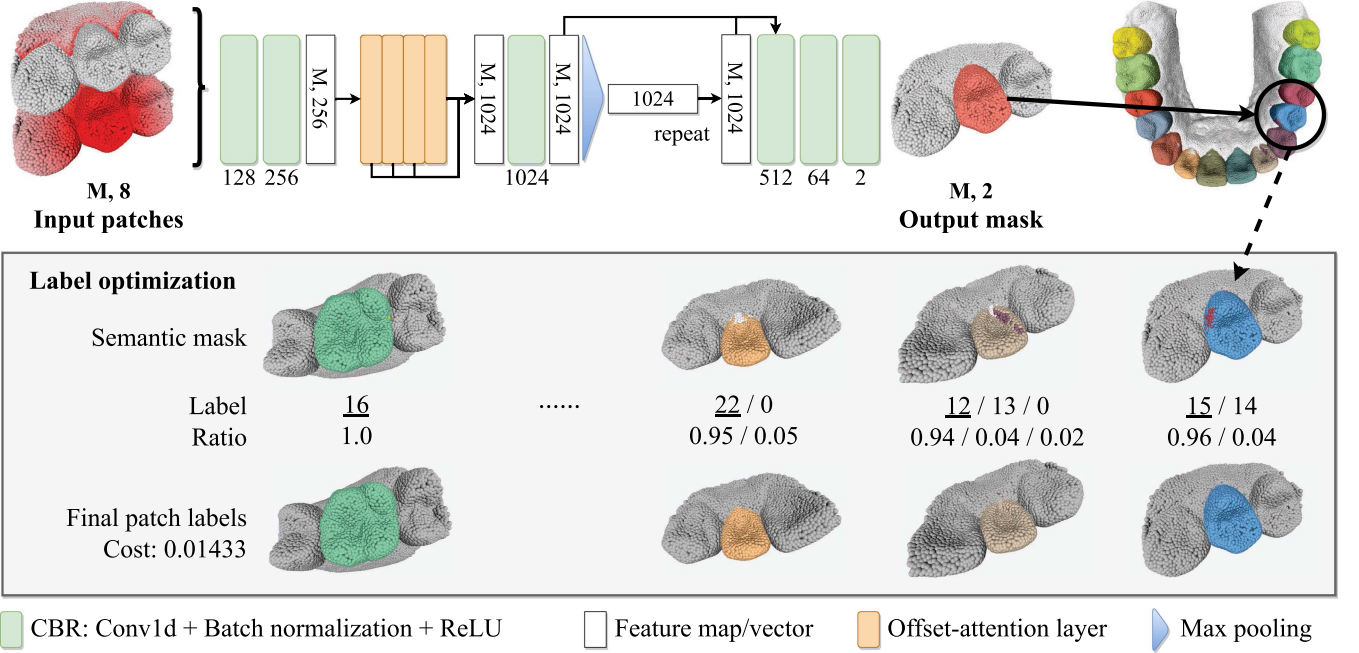


Fig. 8. Network structure of RefineNet for patch segmentation and the illustration for label optimization. The network takes $M \times 6$ patch point clouds with 1-D curvature f_{s_i} and 1-D distance weight w_{s_i} as input, and outputs a mask for each patch. Then, we obtain possible labels and ratios by masking the semantic segmentation result of the dental model for each patch. Finally, after calculating a loss for each label combination, we select the best label combination as the final patch label. The underlined id in this figure is the best label for the corresponding patch.

$$f_{p_i}^c = \frac{1}{k} \sum_{j=1}^k \arccos(\mathbf{n}_i \cdot \mathbf{n}_j), \quad (10)$$

where $\mathcal{N}(\mathbf{p}_i)$ represents the k points closest to the i -th point $\mathbf{p}_i$, k refers to the number of neighbor points, $\hat{l}_{p_j}^b$ is the label of j -th point $\mathbf{p}_j$ in $\mathcal{N}(\mathbf{p}_i)$, and $\mathbf{n}_i, \mathbf{n}_j$ refers to the unit normal vector of point i, j respectively. We express the curvature by the normal changing rate in the neighborhood of the point for computational efficiency.

As shown in Fig. 8, we crop the nearest 4096 points from $\mathbf{P}$ around each centroid as a patch $\mathbf{S}_i$ which contains at least one full tooth, where $0 < i < |\mathbf{C}|$ is the patch index. All the patches form the set $\{\mathbf{S}\}$. To mark the tooth that need to be segmented, a distance weight w_{s_i} is calculated for each point $s_i \in \mathbf{S}_i$ as:

$$w_{s_i} = e^{-2 \times \|\mathbf{s}_i - \hat{\mathbf{c}}_i\|_2}, \quad (11)$$

where $\hat{\mathbf{c}}_i$ is the corresponding centroid of $\mathbf{S}_i$. The point in one patch is composed of $[x, y, z, n_x, n_y, n_z, f_{s_i}, w_{s_i}]$. We use a transformer block with an MLP on $\mathbf{S}_i$ that extracts features and outputs the point-wise confidence $\hat{s}_j$ between 0 and 1. The output mask is supervised by a cross-entropy loss, defined as:

$$\mathcal{L}_{seg} = -\frac{1}{|\mathbf{S}_i|} \sum_{j=1}^{|\mathbf{S}_i|} 1 \cdot \log \hat{s}_j, \quad (12)$$

the total loss function for instance segmentation $\mathcal{L}_{ins}$ is defined as:

$$\mathcal{L}_{ins} = \mathcal{L}_{ct} + \mathcal{L}_{seg}. \quad (13)$$

Finally, we detect patches that overlap with each other and merge them into one patch if their IoU (intersection over union) value is greater than a given threshold, which is set to 0.3 in our experiments.

E. Labeling Optimization and Error Correction

After getting predicted tooth masks in the patches, we mask the semantic segmentation result and obtain possible labels and ratios. The ratio for one label in the patch is the proportion of the number of points with that label to the number of masked points. We have found that directly choosing the label with the maximum ratio is not a good policy, where detailed information is presented in the ablation study. So we propose a cost-ranking algorithm to find the best label sequence. Here are some basic rules:

- 1) The semantic segmentation result is a great prior of teeth distribution which is reliable most of the time.
- 2) No repeated labels are expected.
- 3) The central incisors should exist if possible, for they are located at the important boundary between left and right.
- 4) The id sequence on the left or right side should be as close as possible. For example, [5, 6, 7] is better than [5, 6, 8].

The detailed algorithm is listed in Algorithm 1. Considering that the labels count on each tooth is small, we can get all label sequences for all teeth by enumeration. Then, we calculate the cost for each label sequence. Finally, we select the sequence with the minimum cost as the result.

Although we have considered a lot, labeling errors are still inevitable actually. For a better labeling result, we propose the

Algorithm 1: Sequence Cost-Ranking Algorithm**Require:** A label array l and ratio array p with size n .**Ensure:** l contains at most 2 quadrants.

```

1:  $cost \leftarrow (1 - p[0..n - 1])/n$ ; {higher ratio}
2:  $cost \leftarrow cost + |l| - |\text{unique}(l)|$ ; {no repeat}
3: for each quadrant  $q$  exists in  $l$  do
4:   Get labels in this quadrant as  $l_q$ ;
5:   if  $(q \times 10 + 1) \notin l_q$  then
6:      $cost \leftarrow cost + 1$ ; {expect central incisors}
7:   end if
8:    $cost \leftarrow cost + \max(0, \max(l_q) - \min(l_q) + 1 - |l_q|)$ ; {expect a close sequence}
9: end for
10: return  $cost$ 

```

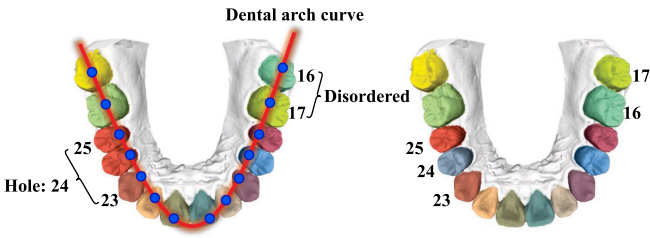


Fig. 9. Label correction postprocessing stage using dental arch curve. The blue dots on the curve represents the nearest point on the curve of each tooth.

postprocessing stage using the dental arch, which is a good prior for teeth arrangement. Since the model is aligned, we project centroids $\hat{C}$ onto xOy plane and fit a para-curve. Then, we obtain the coordinate $(x_{\hat{c}_i}, y_{\hat{c}_i})$ of the nearest point of $\hat{c}_i$ on the curve. Here are some typical labeling errors that usually occur: (1) label disordered; (2) different teeth with the same label; (3) missing teeth.

As shown in Fig. 9, the blue dots on the curve represents the nearest point on the curve of each tooth, which helps determine the relative position between teeth. For case (1), it can be solved by swapping teeth labels based on the order of $x_{\hat{c}_i}$. For case (2), we adopt a hole-filling method. A “hole” is defined as a set $\{x | x \notin \hat{l}\}$, where $\hat{l}$ represents all predicted labels. The method rearranges the labels from the repeated ones until they are well-ordered. Case (3) is more difficult to solve because of labeling ambiguities. For example, if a patient pulls out one premolar, it is difficult to determine the id of the other premolar as the 5th or 6th due to shape similarity. We choose to trust the algorithm’s output while applying the strategies of (1) and (2) to deal with other errors.

IV. EXPERIMENTS AND RESULTS

A. Dataset

We use the Teeth3DS dataset collected by Ben-Hamadou et al. [12] for training and testing. This dataset contains the upper and lower jaw models of 600 subjects, that is, 1200 oral scan models in total. It contains many abnormal cases, including incomplete or missing teeth, crowded teeth, different types

of bases and dental attachments, etc. These models are well labeled by experienced experts. We use 960 of them for training, 120 for validation, and 120 for testing. The input meshes are down-sampled to 32768 points to reduce computational costs and maintain sufficient geometric details.

B. Evaluation Metrics

To better evaluate an algorithm, we use the following metrics to quantitatively measure the results.

Dice similarity coefficient (teeth-gingival): is capable of evaluating the teeth-gingival segmentation quality. It can be calculated by the following formula:

$$DSC_{tg} = \frac{2 \times |pred \cap gt|}{|pred| \cup |gt|} \quad (14)$$

Teeth localization accuracy: is used to measure the distance from a ground truth tooth centroid to its nearest predicted tooth centroid. The distance d is normalized by the ground truth tooth’s size. The accuracy is calculated by $\exp(-d)$. The size of a tooth is defined by the diagonal length of its oriented bounding box.

Due to the reason that teeth labeling and localization accuracy are calculated on each ground truth tooth, we collect the scores of all teeth over the dataset, and take the average value as the result. These three metrics together can evaluate not only the overall segmentation quality but also the ability to locate and classify an individual tooth.

Teeth labeling accuracy: for each ground truth tooth, we calculate a dice coefficient between it and the predicted tooth with the same label. If no predicted tooth with that label is present for any ground truth label, then the prediction accuracy value for this ground truth tooth is set to 0. This metric is the most strict one but is able to reflect the comprehensive ability of accurate classification and segmentation.

A total score is obtained by an average over dice similarity coefficient, teeth localization accuracy, and teeth labeling accuracy. This score is able to measure the segmentation and labeling quality.

Practical usability: is designed to measure the labeling quality of a dental model from the practical application perspective. For each ground truth tooth on a dental model, if its nearest predicted tooth has the same label as it, and the distance from the ground truth centroid to the predicted centroid is under half of the tooth’s size, then we consider the predicted tooth as a true prediction. If every ground truth tooth has a truly predicted tooth on one dental model, then the practical usability is marked as 1, else 0. A dental model with practical usability of 1 can be considered automatically processed; otherwise, it must be manually intervened to correct errors, which dramatically affects the efficiency of the orthodontic process.

C. Comparisons

We compare our method with some representative algorithms in 3D segmentation, including PointNet++ [8], Point Cloud Transformer [23], MeshSegNet [9], TSegNet [11] and

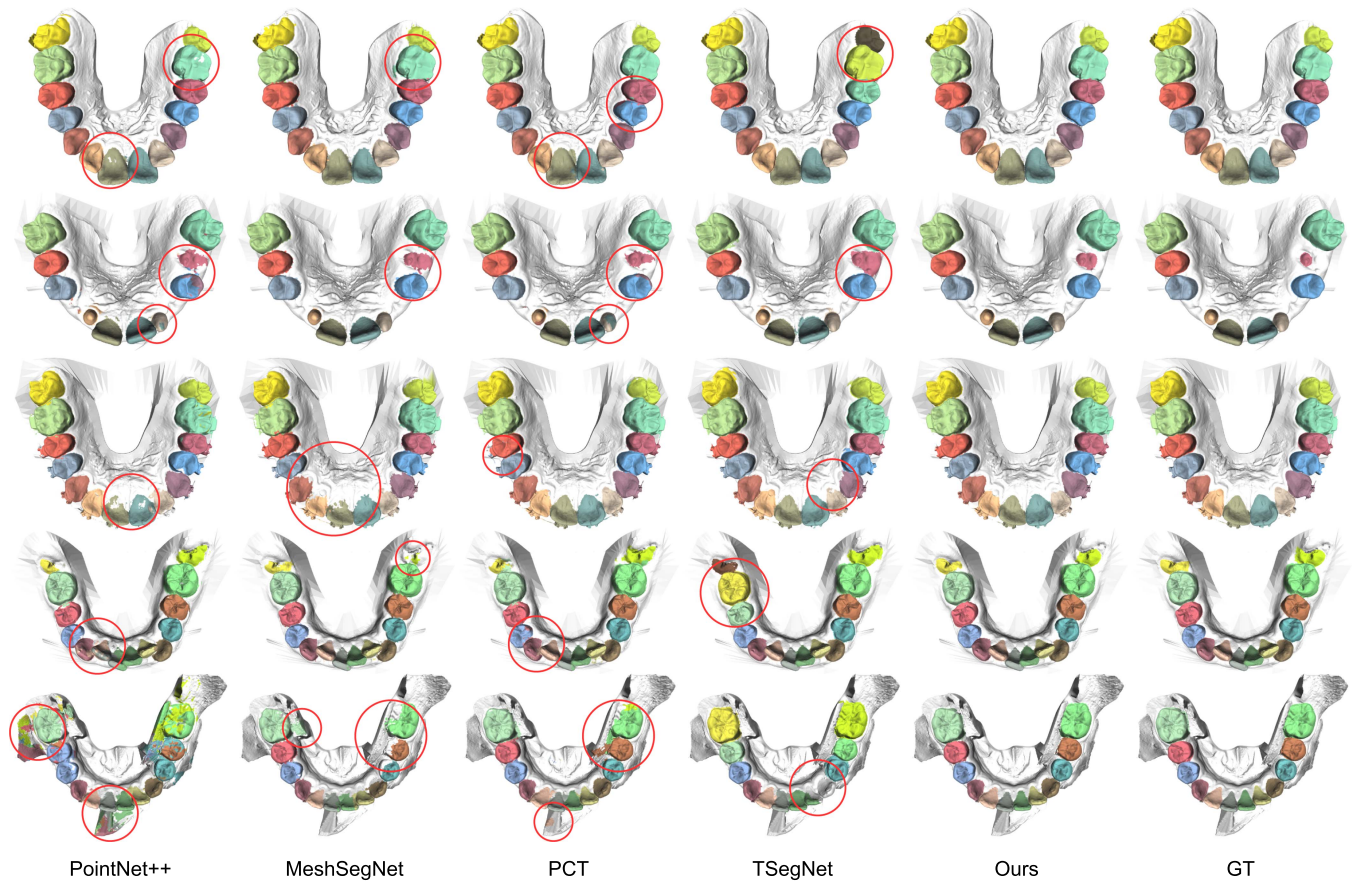


Fig. 10. Visualization results of compared methods and ground truth. The 1st row shows a regular upper jaw model; the 2nd row shows an upper jaw model with small and absent teeth; the 3rd row shows an upper jaw model with dental attachments; the 4th row shows a lower jaw model with crowded teeth; the last row shows a lower jaw model with an interfering base.

SRF-Net [38], using the metrics mentioned above. The first and second algorithms take point clouds as input and achieved state-of-the-art performance on many public datasets, while the other three methods are specialized in solving oral scan segmentation problems. MeshSegNet [9] requires the same number of triangles for each input dental model, so we first down-sampled all the input meshes to 16000 faces with the help of PyMeshLab [42] and trained for 200 epochs. SRF-Net [38] focuses on teeth labeling rather than segmentation, so we only compare our teeth labeling accuracy with it.

We use the semantic segmentation code of PointNet++ and Point Cloud Transformer, and the output channel is set to 33 (32 teeth classes plus one gingiva background). The default output channel for MeshSegNet is 15, which doesn't take the third molars into account, so we change it to 33, which is identical to our method.

We trained all the algorithms to converge, and evaluated their outputs on the testing set. The visualization result is demonstrated in Fig. 10 and the quantitative evaluation is shown in Table I. It can be seen that our method achieves state-of-the-art performance. With the help of the proposed classification method and the postprocessing stage, we achieve better improvements in teeth labeling accuracy, especially.

TABLE I
THE DICE SIMILARITY COEFFICIENT (DSC), TEETH LOCALIZATION ACCURACY (LOC.), TEETH LABELING ACCURACY (LBL-T), THE TOTAL SCORE, AND PRACTICAL USABILITY (LBL-P) OF COMPARISON ALGORITHMS

Method	DSC	Loc.	Lbl-T	Score	Lbl-P
PointNet++ [8]	94.74 [†]	96.82 [†]	89.14 [†]	93.57 [†]	64.17
PCT [23]	96.02 [†]	97.79 [†]	91.21 [†]	95.01 [†]	61.67
MeshSegNet [9]	93.87 [†]	98.36 [†]	90.97 [†]	94.40 [†]	77.50
TSegNet [11]	96.67 [†]	98.67	85.35 [†]	93.56 [†]	50.83
‡SRF-Net [38]	-	-	93.86	-	-
Ours	97.63	99.42	95.62	97.55	90.83

[†] indicates the improvement of our method over other algorithms is significant at the level of 0.01. The ‡ means this method only focuses on labeling, the other metrics are not collected. (%)

Oral scan models acquired by different devices usually have different poses. Even for a set of upper and lower jaw models in an occlusal state, their poses are also different. An ideal solution should not require manual adjustments to the pose, as that would affect orthodontic efficiency. To further evaluate the robustness of our method to arbitrary model rotations, we make a randomly generated rotation, which we believe is representative enough to check the performance of a method. We first align the models

TABLE II
AFTER ARBITRARY ROTATIONS, THE DICE SIMILARITY COEFFICIENT (DSC), TEETH LOCALIZATION ACCURACY (LOC.), TEETH LABELING ACCURACY (LBL-T), THE TOTAL SCORE, AND THE PRACTICAL USABILITY (LBL-P) OF COMPARISON ALGORITHMS

Pre-align	Method	DSC	Loc.	Lbl-T	Score	Lbl-P
PCA	PointNet++ [8]	91.44 (5.52) [†]	88.51 (10.32) [†]	46.13 (35.98) [†]	75.36 (15.44) [†]	14.17
	PCT [23]	91.49 (7.32) [†]	89.49 (11.17) [†]	51.05 (36.79) [†]	77.34 (16.70) [†]	15.83
	MeshSegNet [9]	93.61 (3.25) [†]	96.14 (5.73) [†]	77.47 (21.96) [†]	89.07 (8.72) [†]	41.67
	TSegNet [11]	87.35 (11.28) [†]	91.47 (13.93) [†]	36.79 (38.15) [†]	71.87 (17.64) [†]	13.33
	Ours	96.51 (1.98)	98.98 (3.59)	88.50 (22.25)	94.67 (7.92)	82.50
ICP	PointNet++ [8]	76.10 (37.36)	82.71 (20.40)	63.02 (38.55) [†]	73.94 (30.71)	35.00
	PCT [23]	79.90 (33.92)	83.23 (18.36)	69.68 (37.99)	77.61 (32.75)	39.17
	MeshSegNet [9]	74.14 (31.48)	83.83 (19.63)	60.10 (40.15) [†]	72.69 (29.10)	37.50
	TSegNet [11]	75.74 (36.87)	84.23 (18.16)	70.97 (39.48)	76.98 (35.85)	49.17
	Ours	78.40 (37.02)	84.91 (19.00)	79.51 (41.54)	80.94 (36.55)	79.17
AlignNet	PointNet++ [8]	95.47 (4.17) [†]	96.50 (5.31) [†]	87.74 (12.43) [†]	93.24 (6.31) [†]	52.50
	PCT [23]	96.70 (1.98) [†]	97.35 (4.85) [†]	89.99 (10.05) [†]	94.68 (4.43) [†]	57.50
	MeshSegNet [9]	94.98 (2.36) [†]	97.80 (3.98) [†]	90.24 (8.78) [†]	94.34 (4.00) [†]	60.83
	TSegNet [11]	95.71 (4.01) [†]	97.99 (5.44) [†]	83.82 (15.64) [†]	92.51 (6.70) [†]	49.17
	Ours	97.62 (1.07)	99.13 (3.58)	94.90 (8.50)	97.22 (3.60)	90.00

[†] indicates the improvement of our method over other algorithms is significant at the level of 0.01. The values in brackets are the standard deviations of the metrics. (%)

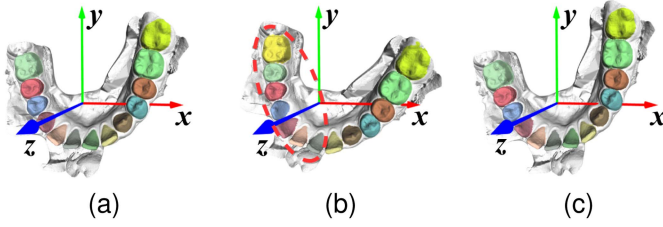


Fig. 11. Visualization result of the aligned models using different aligning methods. (a) Ground truth pose and labels. (b) Aligned using PCA. The teeth in the red circle are mislabeled. (c) Aligned using our AlignNet.

using PCA [39], ICP [40], and our AlignNet respectively, and make evaluations. The results are shown in Table II. It is observed that although the traditional pre-align methods could align the models to a good pose as much as possible, they are still greatly affected by the quality of the models, especially when the models are left-right asymmetric. Our method performs much better when the models are not well aligned.

D. Ablation Study

AlignNet: When given a dental model with an uncertain pose, the easiest way is to find its principal components to determine the pose of the model. However, such methods usually give a bad result when the model is left-right asymmetric. We conduct an experiment to align arbitrarily rotated models using PCA [39], ICP [40] and AlignNet, respectively, and then we compare the angle errors of d'_{up} and d'_{fw} , the two representative directions of dental models, with ground truth directions d_{up} and d_{fw} . As shown in Fig. 11 and Table III, the aligned models using traditional methods have greater angle errors for both d_{up} and d_{fw} , and thus affects the network predictions, especially for teeth labeling where the position features are heavily affected.

DiffLoss and label optimization strategy: The DiffLoss is designed for a better semantic segmentation result, and we add a coefficient μ to balance its weight. We study the influence when

TABLE III
COMPARISON BETWEEN PCA, ICP, AND OUR ALIGNNET ON ANGLE ERRORS AFTER ARBITRARY ROTATIONS. (%)

Method	d_{up} errors		d_{fw} errors	
	Mean	Max	Mean	Max
PCA [39]	7.96°	17.19°	10.99°	24.96°
ICP	14.32°	87.77°	10.67°	73.28°
AlignNet	5.21°	17.18°	5.98°	16.44°

TABLE IV
ABLATION STUDY ON THE SIZE OF DOWN-SAMPLED POINTS IN CENTROIDNET

Size	DSC	Loc.	Lbl-T	Score	Lbl-P
256	97.12 (1.70)	98.32 (6.00)	93.38 (8.46)	96.27 (3.99)	73.33
512	97.63 (1.10)	99.42 (1.03)	95.62 (5.76)	97.55 (2.08)	90.83
1024	97.57 (1.26)	99.31 (2.02)	95.08 (6.80)	97.32 (2.56)	89.17
2048	97.56 (1.35)	99.23 (2.68)	95.31 (6.22)	97.37 (2.52)	90.00

The values in brackets are the standard deviations of the metrics. (%)

setting different values of μ . When μ is set to 0, it is equivalent to removing the $\mathcal{L}_{diff}$ from SemNet, which is able to verify the effectiveness of our proposed loss function.

As shown in Table V, we find that when μ is set to 0.2, the overall accuracy is the highest, which represents that the network produces a better semantic segmentation result. Furthermore, the label optimization algorithm and postprocessing stage enhance the final performance.

Different down-sampling sizes in CentroidNet: We experiment to find the best centroid size in CentroidNet. As shown in Table IV, when the size is set to 512, we gain the best performance. Fewer dense prediction centroids around a small tooth may be seen as noises and won't be clustered as a centroid. More centroids will make the network convergence more difficult and generate more interference points.

Teeth-gingival separation and curvature: The teeth-gingival separation process is designed to eliminate unnecessary parts from dental models. To verify its effectiveness, we conduct an experiment where this operation is removed from our pipeline,

TABLE V
COMPARISON RESULTS WHEN USING DIFFERENT DIFFLOSS WEIGHT (μ)

μ	OA	DSC	Loc.	Lbl-T	Score	Lbl-P
0	94.16 (4.68)	97.63 (1.10)	99.42 (1.03)	94.84 (7.47)	97.30 (2.64)	88.33
0.1	94.11 (5.09)	97.63 (1.10)	99.42 (1.03)	94.39 (11.02)	97.15 (3.77)	90.00
0.2	94.41 (4.09)	97.63 (1.10)	99.42 (1.03)	95.62 (5.76)	97.55 (2.08)	90.83
0.5	94.09 (4.68)	97.62 (1.10)	99.34 (2.13)	94.96 (7.05)	97.31 (2.56)	88.33
1.0	93.33 (5.68)	97.62 (1.10)	99.34 (2.06)	92.83 (14.79)	96.60 (4.99)	86.67

We add an overall accuracy (OA) to evaluate the semantic segmentation quality. the values in brackets are the standard deviations of the metrics. (%)

TABLE VI
COMPARISON RESULTS WHEN USING DIFFERENT LABEL OPTIMIZATION STRATEGIES (STRATEGY), AND POSTPROCESSING OPTIONS (POST.)

Strategy	Post.	DSC	Loc.	Lbl-T	Score	Lbl-P
max	✓	97.63 (1.10)	99.42 (1.03)	85.82 (23.28)	94.29 (7.84)	66.67
max			94.21 (9.71)	74.34 (18.00)	88.73 (7.24)	14.17
best			99.37 (1.48)	95.60 (5.50)	97.53 (2.08)	90.00
best	✓		99.42 (1.03)	95.62 (5.76)	97.55 (2.08)	90.83

We add an overall accuracy (OA) to evaluate the semantic segmentation quality. The “best” strategy means the method ranks all label sequences and finds the best as the result, while “max” means it chooses the predicted label with the most occurrences as the result. The values in brackets are the standard deviations of the metrics. (%)

TABLE VII
ABLATION STUDY OF THE EFFECTIVENESS OF TEETH-GINGIVAL BINARY CLASSIFICATION (BIN-CLS.), CURVATURE (CURV.), AND LABEL CORRECTION POSTPROCESSING (POST.) STAGE

Bin-Cls.	Curv.	Post.	DSC	Loc.	Lbl-T	Score	Lbl-P
✓	✓		97.63 (1.10)	99.37 (1.48)	95.60 (5.50)	97.53 (2.08)	90.00
✓		✓	97.38 (1.14)	99.25 (1.11)	95.27 (5.71)	97.30 (2.06)	90.83
		✓	96.95 (1.42)	98.66 (4.02)	94.26 (7.33)	96.62 (3.28)	86.67
✓	✓	✓	97.63 (1.10)	99.42 (1.04)	95.62 (5.76)	97.55 (2.08)	90.83

The values in brackets are the standard deviations of the metrics. (%)

TABLE VIII
COMPARISON RESULTS WHEN USING DIFFERENT PATCH SIZES

Size	DSC	Loc.	Lbl-T	Score	Lbl-P
3072	96.60 (1.33)	98.98 (1.44)	93.90 (6.22)	96.49 (2.33)	86.67
4096	97.63 (1.10)	99.42 (1.03)	95.62 (5.76)	97.55 (2.08)	90.83
6144	93.78 (2.94)	96.66 (4.90)	86.56 (10.50)	92.33 (4.88)	51.67

The values in brackets are the standard deviations of the metrics. (%)

and we analyze the resulting performance on the testing dataset. Additionally, as curvature information on teeth crowns needs to be removed based on the binary classification result, we will not include curvature information in RefineNet when the teeth-gingival separation process is omitted. The comparative results are presented in Table VII. Experimental results show that this strategy can affect the prediction of centroids. If this operation is omitted, small teeth can be overlooked easily. Similarly, removing curvature information results in less distinct boundaries for segmentation. At times, the area labeled as teeth may significantly deviate from the ground truth at the boundaries, as shown in Fig. 12.

Patch size: The patch size plays a key role in the single tooth segmentation. A large size of patch contains more teeth, leading to the difficulty in distinguishing teeth to be segmented, while a small size of patch is hard to fully include larger teeth such as molars. We compare the results using different patch sizes, and the results are shown in Table VIII. Based on the experimental testings, we choose 4096 as the patch size.

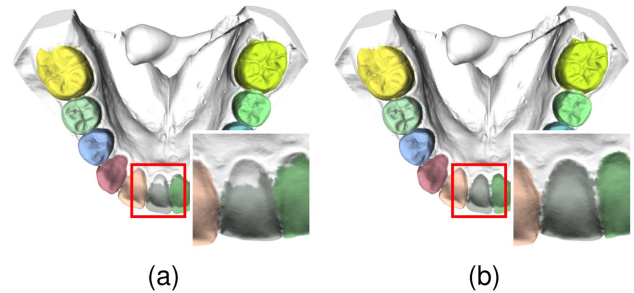


Fig. 12. Visualization result on curvature in single tooth segmentation. (a) Without curvature. (b) With curvature.

Postprocessing: We verify the effectiveness of our label correction process and different label optimization strategies. The results are shown in Tables VI and VII. When the postprocessing stage is removed, the overall segmentation area on the whole model won't be affected. However, the possibility of label disordered or label repeated will heavily affect the performance. Without finding the best label sequence, the predicted labels will be determined mainly by the quality of SemNet results.

E. Limitation and Future Work

Since we make predictions on down-sampled point clouds, many points on the boundary between teeth and gingiva are lost. When predicted labels are interpolated back to the original models, the teeth segmentation boundaries are not smooth enough.

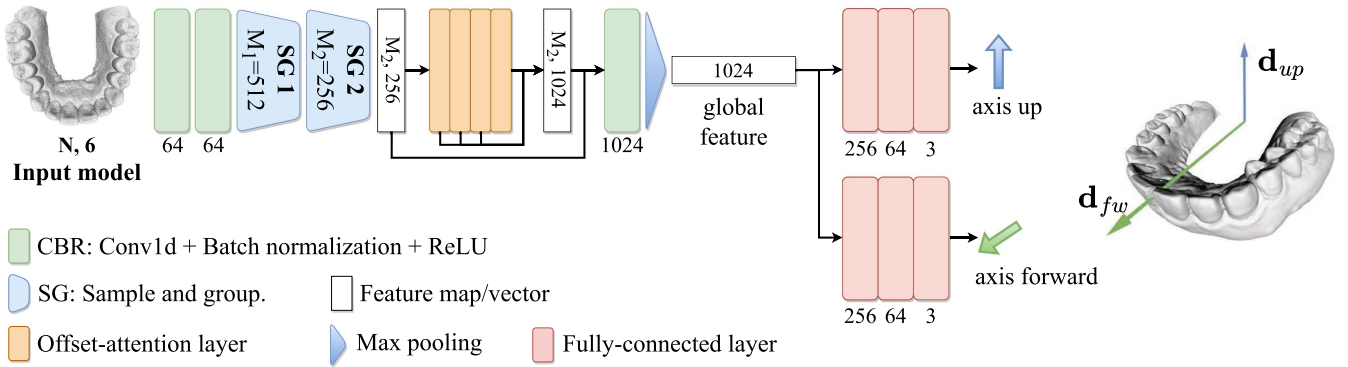


Fig. 13. Detailed AlignNet structure.

We will work on smoothing teeth-gingival boundaries in future works.

V. CONCLUSION

In this work, we propose a robust teeth segmentation and labeling system for intraoral scan models. First, we design an AlignNet to align models with different poses from different devices. Then we combine the advantages of semantic segmentation and instance segmentation to improve the accuracy of teeth segmentation and classification. For the teeth-gingival weak boundaries, we use prior information (curvature) as a guide to further improve segmentation accuracy. Meanwhile, a label optimization and error correction strategy are proposed to further improve the classification accuracy. For different evaluation metrics (especially the practical usability), our method outperforms state-of-the-art methods for teeth segmentation and labeling on the Tooth3DS dataset.

APPENDIX A EXPERIMENT DETAILS

A. Network Parameters

We have shown parts of network parameters in the network structure figures. The number below the rectangle (the CBR block or the fully-connected layer) is the size of the output channel, and the text inside the rectangle means the size of the feature vector (or feature map). In this section, we will mainly illustrate the parameters that are not shown in the figures.

AlignNet: The AlignNet is almost identical to Point Cloud Transformer [23]. As shown in Fig. 13, we use 2 sampling and grouping (SG) blocks to down-sample the input cloud. An SG block down-samples the input point cloud and aggregates local neighborhood features. For the first block ‘SG1’, the KNN radius $r=0.15$, neighbors count $k=32$, the output channels of MLPs are (128, 128). Similarly, for ‘SG2’, the radius $r=0.2$, neighbors count $k=32$, and the output channels of MLPs are (256, 256). The input and output channel dimensions of the offset attention layers are 256.

SemNet: As shown in Fig. 4, we use 4 multi-scale sampling and grouping (MSG) blocks with 4 feature propagation (FP) blocks for point-wise semantic segmentation. An MSG block is composed of several SG blocks, and output features of SG are

concatenated together. The parameters for an MSG block are arranged in brackets in the order of SG blocks. An FP block up-samples the input point cloud and features using 3-NN interpolation with MLPs. For ‘MSG1’, $r=[0.025, 0.05]$, $k=[16, 32]$, MLPs are [(16, 16, 32), (32, 32, 64)]. For ‘MSG2’, $r=[0.05, 0.1]$, $k=[16, 32]$, MLPs are [(64, 64, 128), (64, 96, 128)]. For ‘MSG3’, $r=[0.1, 0.2]$, $k=[16, 32]$, MLPs are [(128, 196, 256), (128, 196, 256)]. For ‘MSG4’, $r=[0.2, 0.4]$, $k=[16, 32]$, MLPs are [(128, 196, 256), (128, 196, 256)]. For the 4 FP blocks, MLPs are (512, 512), (512, 512), (256, 256) and (128, 128) respectively. The output feature map of ‘MSG4’ is fed into 4 offset attention layers, concatenated to form a 64×2048 feature map, and fed into a (2048, 1024, 512, 512) shared MLP, followed by a max pooling operation. The 512-d global feature vector is repeated N times and concatenated with the 128-d features from ‘FP4’, and fed into a (640, 256, 33) shared MLP whose output is the 33-d pointwise label.

CentroidNet: As shown in Fig. 5, we use 3 sampling and grouping blocks in this network. For ‘SG1’, $r=0.025$, $k=32$, MLPs are (128, 128). For ‘SG2’, $r=0.05$, $k=32$, MLPs are (256, 256). For ‘SG3’, $r=0.1$, $k=16$, MLPs are (512, 512). The input and output channel dimensions of offset attention layer are 512.

REFERENCES

- [1] T. Yuan, W. Liao, N. Dai, X. Cheng, and Q. Yu, “Single-tooth modeling for 3D dental model,” *Int. J. Biomed. Imag.*, vol. 2010, 2010, Art. no. 535329.
- [2] K. Wu, L. Chen, J. Li, and Y. Zhou, “Tooth segmentation on dental meshes using morphologic skeleton,” *Comput. Graph.*, vol. 38, pp. 199–211, 2014.
- [3] C. Sinthanayothin and W. Tharanont, “Orthodontics treatment simulation by teeth segmentation and setup,” in *Proc. IEEE 5th Int. Conf. Elect. Eng./Electron., Comput., Telecommun. Inf. Technol.*, 2008, pp. 81–84.
- [4] M. Yaqi and L. Zhongke, “Computer aided orthodontics treatment by virtual segmentation and adjustment,” in *Proc. IEEE Int. Conf. Image Anal. Signal Process.*, 2010, pp. 336–339.
- [5] J. Long, E. Shelhamer, and T. Darrell, “Fully convolutional networks for semantic segmentation,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 3431–3440.
- [6] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. Learn. Representations*, 2021.
- [7] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, “PointNet: Deep learning on point sets for 3D classification and segmentation,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 652–660.
- [8] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, “PointNet++ : Deep hierarchical feature learning on point sets in a metric space,” in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 5105–5114.

- [9] C. Lian et al., "Deep multi-scale mesh feature learning for automated labeling of raw dental surfaces from 3D intraoral scanners," *IEEE Trans. Med. Imag.*, vol. 39, no. 7, pp. 2440–2450, Jul. 2020.
- [10] F. G. Zanjani et al., "Mask-MCNet: Tooth instance segmentation in 3D point clouds of intra-oral scans," *Neurocomputing*, vol. 453, pp. 286–298, 2021.
- [11] Z. Cui et al., "TSegNet: An efficient and accurate tooth segmentation network on 3D dental model," *Med. Image Anal.*, vol. 69, 2021, Art. no. 101949.
- [12] A. Ben-Hamadou et al., "Teeth3DS: A benchmark for teeth segmentation and labeling from intra-oral 3D scans," 2022, *arXiv:2210.06094*.
- [13] Y. Li et al., "PointCNN: Convolution on x-transformed points," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 828–838.
- [14] Y. Wang et al., "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, pp. 1–12, 2019.
- [15] W. Wu, Z. Qi, and L. Fuxin, "PointConv: Deep convolutional networks on 3D point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9621–9630.
- [16] H. Thomas et al., "KPConv: Flexible and deformable convolution for point clouds," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 6411–6420.
- [17] F. Engelmann, T. Kontogianni, and B. Leibe, "Dilated point convolutions: On the receptive field size of point convolutions on 3D point clouds," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 9463–9469.
- [18] L. Hu, M. Qin, F. Zhang, Z. Du, and R. Liu, "RSCNN: A CNN-based method to enhance low-light remote-sensing images," *Remote Sens.*, vol. 13, no. 1, 2020, Art. no. 62.
- [19] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.
- [20] N. Carion et al., "End-to-end object detection with transformers," in *Proc. 16th Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.
- [21] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 6881–6890.
- [22] H. Zhao, L. Jiang, J. Jia, P. H. Torr, and V. Koltun, "Point transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 16259–16268.
- [23] M.-H. Guo et al., "PCT: Point cloud transformer," *Comput. Vis. Media*, vol. 7, pp. 187–199, 2021.
- [24] M. Zhao, L. Ma, W. Tan, and D. Nie, "Interactive tooth segmentation of dental models," in *Proc. IEEE Eng. Med. Biol. 27th Annu. Conf.*, 2005, pp. 654–657.
- [25] Y. Kumar, R. Janardan, B. Larson, and J. Moon, "Improved segmentation of teeth in dental models," *Comput.-Aided Des. Appl.*, vol. 8, no. 2, pp. 211–224, 2011.
- [26] Z. Li and H. Wang, "Interactive tooth separation from dental model using segmentation field," *PLoS one*, vol. 11, no. 8, 2016, Art. no. e0161159.
- [27] S. Tian et al., "Automatic classification and segmentation of teeth on 3D dental model using hierarchical deep learning networks," *IEEE Access*, vol. 7, pp. 84817–84828, 2019.
- [28] Y. Zhao et al., "TSASNet: Tooth segmentation on dental panoramic x-ray images by two-stage attention segmentation network," *Knowl.-Based Syst.*, vol. 206, 2020, Art. no. 106338.
- [29] A. Poonsri, N. Aimjirakul, T. Charoenpong, and C. Sukjamsri, "Teeth segmentation from dental X-ray image by template matching," in *Proc. IEEE 9th Biomed. Eng. Int. Conf.*, 2016, pp. 1–4.
- [30] X. Xu, C. Liu, and Y. Zheng, "3D tooth segmentation and labeling using deep convolutional neural networks," *IEEE Trans. Visual. Comput. Graph.*, vol. 25, no. 7, pp. 2336–2348, Jul. 2019.
- [31] L. Zhang et al., "TSGCNet: Discriminative geometric feature learning with two-stream graph convolutional network for 3D dental model segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 6699–6708.
- [32] Y. Zheng, B. Chen, Y. Shen, and K. Shen, "TeethGNN: Semantic 3D teeth segmentation with graph neural networks," *IEEE Trans. Vis. Comput. Graph.*, vol. 29, no. 7, pp. 3158–3168, Jul. 2023.
- [33] D. Sun et al., "Tooth segmentation and labeling from digital dental casts," in *Proc. IEEE 17th Int. Symp. Biomed. Imag.*, 2020, pp. 669–673.
- [34] Y. Tian et al., "3D tooth instance segmentation learning objectness and affinity in point cloud," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 18, no. 4, pp. 1–16, 2022.
- [35] D. Liu, Y. Tian, Y. Zhang, J. Gelernter, and X. Wang, "Heterogeneous data fusion and loss function design for tooth point cloud segmentation," *Neural Comput. Appl.*, vol. 34, no. 20, pp. 17371–17380, 2022.
- [36] J. Im et al., "Accuracy and efficiency of automatic tooth segmentation in digital dental models using deep learning," *Sci. Rep.*, vol. 12, no. 1, 2022, Art. no. 9429.
- [37] P. Lalitha, M. Gayathri, L. Tamilselvi, and A. Arun, "Application of graceful labeling in dental arch," *Drug Innov. Today*, vol. 11, no. 3, pp. 637–638, 2019.
- [38] Q. Ma et al., "SRF-Net: Spatial relationship feature network for tooth point cloud classification," in *Comput. Graph. Forum*, vol. 39, no. 7. Hoboken, NJ, USA: Wiley, 2020, pp. 267–277.
- [39] L. I. Smith, "A Tutorial on Principal Components Analysis," Dept. Comput. Sci., Univ. Otago, Dunedin, New Zealand, Tech. Rep. OUCS-2002-12, 2002.
- [40] P. J. Besl and N. D. McKay, "Method for registration of 3-D shapes," in *Proc. SPIE*, vol. 1611, pp. 586–606, 1992.
- [41] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, 2014.
- [42] A. Muntoni and P. Cignoni, "PyMeshLab v2022.2.post2," Mar. 2022. [Online]. Available: <https://doi.org/10.5281/zenodo.6337478>



Shaojie Zhuang received the B.S. degree in 2021 from the School of Software, Shandong University, Jinan, China, where he is currently working toward the Ph.D. degree. His research interests include machine learning, medical image processing, and geometric analysis.



Guangshun Wei received the B.S. degree in computer science and technology from the College of Information Science and Engineering, University of Jinan, Jinan, China, in 2017. He is currently working toward the Ph.D. degree with the School of Software, Shandong University, Jinan. His research interests include machine learning, image processing, and geometric analysis.



Zhiming Cui received the Ph.D. degree from the Department of Computer Science, The University of Hong Kong, Hong Kong, in 2022. He is currently an Assistant Professor with the School of Biomedical Engineering, ShanghaiTech University, Shanghai, China. His research interests include medical image analysis, deep learning, and 3D vision. His current research interests include advanced medical imaging computing algorithms, digital dentistry, and medical image reconstruction.



Yuanfeng Zhou received the master's and Ph.D. degrees from the School of Computer Science and Technology, Shandong University, Jinan, China, in 2005 and 2009, respectively. From 2009 to 2011, he held a Postdoctoral position with the Graphics Group, Department of Computer Science, The University of Hong Kong, Hong Kong. He is currently a Professor with the School of Software, Shandong University, where he is also the Leader of the IGIP Laboratory. His research interests include geometric modeling, information visualization, and image processing.